Name: _____

Class: _____

ACTIVITY SHEET

4.1 Food chains and food webs

Each unique ecosystem is home to communities of organisms that interact with each other. While the species in each ecosystem differ, there are some similarities to be found throughout all systems. One of these is energy flow. Scientists use food chains and food webs to demonstrate the flow of energy through an ecosystem.

1 Define the following terms.

a Producer

b Herbivore

c Consumer

2 **a** Write a simple food chain found in the Pacific Ocean.

b All food chains start with a producer. Explain why they do not start with the Sun.

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3 a Read the information below and guess: Who am I?

'Sitting high up in the canopy on the edge of a bromeliad pond, I survey my surrounds in the early morning. It was getting hot and humid and I was warming up nicely. I watch slugs and snails feed off some algae along the pond's edge. Mosquitos come to lay their eggs in the pond. A dragonfly hovers above the pond and starts feeding upon the insect larvae. A small green tree frog snaps its tongue and gets its first feed of the day. Focused on the dragonfly, the frog does not see me coming.'

b Write a food chain based on the information.

c Suggest an ecosystem in which this food chain could be found.

4 a Create a food web with the following organisms:

wombat, termites, magpie, wattle, crickets, native grass, emu, wattle bird, kangaroo, blue-tongued lizard, kookaburra.

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b State the trophic level of each of the bird species.

5 a What is the 10% rule in ecology?

b Starting with a producer that creates 2000 units of energy, create a flowchart showing the units of energy passed on to the next three trophic levels.

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c Create a mind map that shows the inputs and outputs of consumers.

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